

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: AIT304**Course Name: ROBOTICS AND INTELLIGENT SYSTEM**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | Briefly describe the Dynamic characteristics of Mobile robot | (3) |
| 2 | Categorise the various types of grippers used in robot manipulators | (3) |
| 3 | Explain Dynamic range, Linearity and Resolution of a sensor | (3) |
| 4 | Compare CCD camera with CMOS Camera. | (3) |
| 5 | Explain the camera sensor hardware interfacing | (3) |
| 6 | Differentiate between holonomic and nonholonomic robots. | (3) |
| 7 | How will you represent the position and orientation of a wheeled mobile robot? | (3) |
| 8 | State the applications and advantages of SLAM robots. | (3) |
| 9 | Explain the Bug algorithm for obstacle avoidance | (3) |
| 10 | What do you mean by control decomposition? | (3) |

PART B*Answer one question from each module, each carries 14 marks.***Module I**

- | | | |
|----|---|-----|
| 11 | a) Explain the general features of wheeled, legged and aerial robots. | (9) |
| | b) Draw and Explain RPP and RRR robot configuration | (5) |

OR

- | | | |
|----|---|------|
| 12 | a) Briefly explain the types of grippers used in robot manipulator. | (10) |
| | b) Assume an object of mass 140 kg is to be lifted up with an acceleration of 10 m/s ² . Calculate the gripper force required for the operation, if coefficient of friction between contact surfaces is 0.2, number of fingers in gripper is 2 and acceleration due to gravity is 9.8 m/s ² | (4) |

Module II

- 13 a) Explain the working principle of Hall effect sensor and any one application of The same. (9)
- b) Illustrate the significance of the PID controller with a neat block diagram. (5)

OR

- 14 a) Explain the working of any two position sensors with neat sketch. (6)
- b) A mobile robot is designed for unidirectional motion with constant velocity. Design a mechanism to make the robot move in forward and reverse directions with controllable speed and suggest the sensors and motion control Drives required achieving the same. (8)

Module III

- 15 a) Outline the seven stages of robot vision (9)
- b) A frame F has been moved 10 units along y axis and 5 units along z axis of reference frame. Find new location of Frame (5)

$$F = \begin{bmatrix} 0.527 & -0.574 & 0.628 & 5 \\ 0.369 & 0.819 & 0.439 & 3 \\ -0.766 & 0 & 0.643 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

OR

- 16 a) What is an omnidirectional robot? Explain two configurations to set up an omnidirectional robot. (6)
- b) What is meant by transformation and represent pure translation ,pure rotation and combined transformation (8)

Module IV

- 17 a) Describe the concept of mobile robot localization with suitable Block diagrams (7)
- b) Explain the challenges of localization (7)

OR

- 18 a) Give the mathematical definition of SLAM and elaborate the concept. (6)
- b) Apply the concept of Accurate Odometry and Error Modelling for a Mobile Robot with a neat sketch. (8)

Module V

- 19 a) Illustrate the working of Dijkstra's algorithm with a suitable example. (7)

- b) Illustrate the Incorporation of Neural network approach in Robot navigation? (7)
List its advantages

OR

- 20 a) Explain Deterministic graph search algorithms in detail (7)
b) With an example, elaborate the process of horizontal decomposition of a robotic control system. (7)
